

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Branch: CSE(AI)

Semester : VII

Subject Code & Name: Blockchain Technology BTAIIIE705B

Max Marks: 60

Date:11-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Compare and contrast peer-to-peer networks with traditional client-server architectures, highlighting the advantages and disadvantages of each.	Understand	6
B) Define the double spend problem and explain why it is a significant concern in digital currency systems.	Synthesize	6
C) Differentiate between public, private, and consortium blockchains.	Understand	6
Q.2 Solve Any Two of the following.		12
A) Explain the application of SHA-256 algorithm in digital signing verification in detail	Applying	6
B) Compare hash pointer and Merkle tree and its application in Block chain security.	Analyzing	6
C) Define ECDSA and explain how it functions as a digital signature algorithm.	Understand	6
Q.3 Solve Any Two of the following.		12
A) Describe the principles behind Proof of Stake (PoS) as a consensus algorithm. How does it differ from Proof of Work?	Understand	6
B) Explain the role of Practical Byzantine Fault Tolerance (PBFT) in achieving consensus in a distributed network.	Understand	6
C) Differentiate between hard forks and soft forks in a blockchain. Provide examples of situations that might lead to each type of fork.	Analyze	6

Q.4 Solve Any Two of the following. 12

- | | | |
|---|------------|---|
| A) Define the term "World State" in the context of Ethereum. How is it different from the blockchain itself? | Analyze | 6 |
| B) Explain the role of the Ethereum Virtual Machine (EVM) in the Ethereum ecosystem. | Understand | 6 |
| C) Define decentralized applications (DApps) and provide examples of how they operate on the Ethereum blockchain. | Understand | 6 |

Q. 5 Solve Any Two of the following. 12

- | | | |
|--|------------|---|
| A) Discuss the role of blockchain in supply chain management. How does it address challenges such as fraud, inefficiencies, and lack of transparency in supply chains? | Understand | 6 |
| B) Describe the potential benefits and challenges of using blockchain for cross-border payments. https://www.batuonline.com | Analyze | 6 |
| C) Discuss Blockchain and Metaverse | Understand | 6 |

*** End ***